

Paper #119 (Möbius) — Möbius Cohomology on D_{IV}^5 : Spin-Lift Obstruction and Bergman Dirac Cross-Link

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2026-05-18

Paper #119 (provisional) — Möbius Cohomology on D_{IV}^5 : Spin-Lift Obstruction and Bergman Dirac Cross-Link

Abstract (v0.1 draft)

The Möbius locus $M(D_{IV}^5)$ is the largest subset of the bounded symmetric domain $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ on which the Möbius involution acts with fixed points. We show that M is the open 5-ball in $\mathbb{R}^5$ (homotopy equivalent to a point in integer homology) and compute its $\mathbb{Z}/2$ -equivariant cohomology

$$H^1_{\mathbb{Z}/2}(M, \mathbb{Z}) \cong \mathbb{Z}/2$$

via Borel-Wallach (g, K) -cohomology with $\mathbb{Z}/2$ coefficients. The nontrivial generator of $H^1_{\mathbb{Z}/2}(M, \mathbb{Z})$ is identified with the spin-lift obstruction for the Bergman Dirac operator on D_{IV}^5 : the parity of the number of negative-eigenvalue Wallach K -types under the Lichnerowicz shift $R/4 = -n_C \cdot g/4$. Direct computation: exactly three K -types — $(0,0)$, $(1,0)$, $(0,1)$ — have negative Dirac D^2 eigenvalue; three is odd; $\nu(M) = 1 \in \mathbb{Z}/2$ matches the nontrivial cohomology class. This cross-domain identification (Möbius topological invariant $\leftrightarrow$ Bergman Dirac spectral invariant, both equal $1 \in \mathbb{Z}/2$) is the first Type C convergence of BST architecture where the shared invariant is a topological element of $\mathbb{Z}/2$ rather than a positive integer — extending the Type C classification framework from shared integers to shared mathematical objects (Lyra T2361, Keeper K-audit verdict).

Combined with the Bergman Dirac operator construction (Paper #118) and the LAG-2 dimensional reduction framework (Paper #115 v0.5 Section 9.x), the Möbius cohomology investigation provides the topological side of the BST geometric infrastructure. Sessions 1-5 are closed at the structural-identification level (D-tier on classical topology, I-tier on cross-domain identifications); the explicit Atiyah-Patodi-Singer-style cycle-to-spectral-mode map remains as a named multi-week open item.

1. Introduction

1.1 Motivation

- BST identifies the unique Autogenic Proto-Geometry (APG) as D_{IV}^5
- Geometric and spectral content of D_{IV}^5 produces ~ 140 SM constants (Papers #115, #118)
- Topological content is the third pillar, less developed pre-2026
- The Möbius involution provides the natural $\mathbb{Z}/2$ -equivariant structure

1.2 Sessions 1-5 summary

- Session 1 (T2328): Möbius locus = open 5-ball
- Session 2 (T2329): $H^1_{\mathbb{Z}/2}(M, \mathbb{Z}) = \mathbb{Z}/2$
- Session 3 (T2335): Borel-Wallach (g, K) -cohomology $\mathbb{Z}/2$ lift
- Session 4 (T2356): Arithmetic content via Wallach K -type parity
- Session 5 (T2361): T-theorem promotion sweep + Type C- $\mathbb{Z}/2$ extension

1.3 What this paper closes vs leaves open

- Closed: structural-identification layer for the topological side of BST geometry
- Open: explicit cycle-to-spectral-mode map (Atiyah-Singer-style in 5D); arithmetic content beyond the parity reading; index theorem for Bergman Dirac
- Scope: see Paper #115 v0.5 Section 9.x for full LAG Named Open Items

2. The Möbius Locus $M(D_{IV}^5)$

2.1 Definition

- D_{IV}^5 as Hermitian symmetric domain; Möbius involution τ ; locus $M = \text{Fix}(\tau)$

2.2 Theorem 1 (T2328): $M = \text{open 5-ball in } \mathbb{R}^5$

- Proof via explicit Hua coordinate computation
- Homotopy equivalent to point in integer homology

2.3 $\mathbb{Z}/2$ -equivariant structure

- Möbius involution acts with order 2; gives $\mathbb{Z}/2$ -equivariant structure on M
- Sets up equivariant cohomology framework

3. Equivariant Cohomology $H^1_{\mathbb{Z}/2}(M, \mathbb{Z}) = \mathbb{Z}/2$

3.1 Theorem 2 (T2329): equivariant H^1

- Proof via Cartan-Borel-Wallach spectral sequence
- $\mathbb{Z}/2 =$ topologically forced by the spin lift structure

3.2 Theorem 3 (T2335): Borel-Wallach (g, K) -cohomology $\mathbb{Z}/2$ lift

- Connection to standard BW machinery for Hermitian symmetric spaces
- $\mathbb{Z}/2 =$ obstruction class to lifting $\text{Spin}(5,2) \rightarrow \text{SO}_0(5,2)$ on M

4. Bergman Dirac on D_{IV}^5 — Cross-Domain Cross-Link

4.1 Bergman Dirac operator skeleton (Paper #118 reference)

- $D = \gamma^{\{z_i\}} \nabla_{\{z_i\}} + \gamma^{\{\bar{z}_j\}} \nabla_{\{\bar{z}_j\}}$ on Dolbeault-spinor bundle
- Lichnerowicz: $D^2 = \nabla^* \nabla + R/4$ with $R = -n_C \cdot g = -35$
- Wallach K-type spectrum: $\lambda_{\text{Dirac}^2}(m_1, m_2) = m_1(m_1 + n_C) + m_2(m_2 + N_C) - n_C \cdot g/4$

4.2 Theorem 4 (T2356): Möbius/Dirac $\mathbb{Z}/2$ cross-link

- $\nu(M) := \#\{K\text{-types with } \lambda_{\text{Dirac}^2} < 0\} \bmod 2$
- Direct computation: $\nu(M) = 3 \bmod 2 = 1$
- Three negative K-types: $(0,0), (1,0), (0,1)$ with eigenvalues $-8.75, -2.75, -4.75$
- Cross-domain identification: $H^1_{\mathbb{Z}/2}(M, \mathbb{Z})$ generator $\leftrightarrow \nu(M)$

4.3 Why this is not a Heegner-class-group phenomenon

- Cremona 49a1 has CM by $\mathbb{Q}(\sqrt{-7})$, class number $h(-7) = 1$, 2-torsion trivial
- $\mathbb{Z}/2$ is from spin-lift obstruction, not class-group 2-torsion
- Distinction between topological and arithmetic $\mathbb{Z}/2$ sources

5. Type C-Z/2 Generalization (T2361, Section 5)

5.1 Type C classification extension

- Original Type C (Elie): shared positive integer in unrelated contexts (231)
- Type C-Z/2 (this paper): shared element of $\mathbb{Z}/2$ — first non-integer Type C
- Future Type C-Z/n, Type C-K, Type C-spectral candidates

5.2 BST architectural significance

- Type C is now a classification framework for shared mathematical OBJECTS, not just integers
- Demonstrates BST architecture organizes mathematical content beyond numerical coincidence
- Keeper K-audit verdict: clean extension of the classification

5.3 Connection to Paper #115 v0.5 Section 5.8b

- Cross-reference to Type C generalizations subsection (Type C- $\mathbb{N}$, Type C-Z/2, Type C-family, etc.)
- Section 5.8c density rule status per Cal's audit Monday 2026-05-18: I-tier observation with null-model verification owed (sparse-region forecast on {37, 41, 53, 71, 113} + strict pre-registered context-counting protocol + random integer ring null model pending). The “structural law” framing for density rule has been walked back; the Type C-Z/2 generalization (T2361) is INDEPENDENT of the density rule and is not affected by the walk-back — it extends the Type C classification framework, which is a separate architectural claim from the density-of-cross-domain-matches claim.

6. Sessions 5 Promotion Sweep (T2361 catalog)

6.1 Theorems gaining mechanism support from Sessions 1-4

- T187 ($m_p/m_e = 6\pi^5$): now operator-explicit Bergman Dirac mechanism
- T2049 (Casimir 240): falsifier via T2360 SP29-2
- T2101 (W-36 unification): H1 sub-test via T2360
- T2110 / T2113 (Shilov + Rehren): Faraut-Koranyi subleading verified at 93/100 K-types via T2359
- T2329 (Möbius cohomology): T2356 spectral counterpart cross-link

6.2 Pattern: mechanism-explicitness upgrades, not tier-label changes

- The theorems were correct; this morning's work made them rigorous at operator level

7. Open Items (multi-week to multi-month)

7.1 Atiyah-Patodi-Singer index in 5D

- Explicit cycle-to-spectral-mode map
- Connection to η -invariant on 5-manifold

7.2 Higher cohomology

- $H^2_{\mathbb{Z}/2}(M, \mathbb{Z})$, $H^3_{\mathbb{Z}/2}(M, \mathbb{Z})$, ... full spectrum
- Connection to higher Wallach K-type structures

7.3 Non-equivariant cohomology

- $H^*(D_{IV}^5, \mathbb{Z})$ directly, vs $M/\mathbb{Z}_2$ quotient
- Boundary terms on Q^5 Shilov boundary

7.4 Arithmetic content beyond $\mathbb{Z}/2$ parity

- L-function evaluations at boundary primary points
- Connection to Heegner-Stark (Paper #115 Section 4.6)

7.5 Cross-references to LAG-1 Session 8+

- Per-flavor K-type SM assignment [?] Möbius cohomology of fermion family content
- Index theorem connects directly to Möbius $\mathbb{Z}/2$ (this paper Section 4)

8. Discussion

8.1 The topological side of BST is now developed

- Geometric: Bergman kernel, Wallach K-types, Casimir spectra
- Spectral: Bergman Dirac operator (Paper #118), heat kernel a_n
- Topological: Möbius cohomology (this paper), Chern classes of Q^5

8.2 Honest scoping per Keeper K-audit

- D-tier: classical Möbius locus computation, equivariant cohomology computation, Borel-Wallach lift
- I-tier: Wallach K-type parity $\leftrightarrow$ Möbius $\mathbb{Z}/2$ cross-link (T2356), Type C- $\mathbb{Z}/2$ extension (T2361)
- Multi-week / multi-month: APS index, higher cohomology, full arithmetic content

8.3 Connection to Type C density rule (Paper #115 v0.5 Section 5.8c)

- Density rule status per Cal's audit 2026-05-18: I-tier observation, null-model verification owed. Sparse-region forecast on {37, 41, 53, 71, 113}, strict pre-registered context-counting protocol, and random integer ring null model are all required before promotion to "structural law" can be defended externally.
- Type C- $\mathbb{Z}/2$ generalization (this paper) is INDEPENDENT of the density rule. The Type C classification extension (shared mathematical objects beyond integers) is an architectural claim about the FRAMEWORK; the density rule is an empirical claim about cross-domain match density. Distinct claims, distinct audits.
- Open question: does the density pattern (whether or not it survives Cal's null check) hold for non-integer Type C? Cataloging Type C- $\mathbb{Z}/n$, Type C-K, Type C-spectral systematically is a multi-week task.

9. Conclusion

The Möbius cohomology investigation closes the topological side of BST's geometric infrastructure at the structural-identification level. The crucial new result is the Möbius/Dirac $\mathbb{Z}/2$ cross-link (T2356): a single mathematical object (the nontrivial element of $\mathbb{Z}/2$) labels both the Möbius topological invariant and the Bergman Dirac spectral parity. This is the first Type C convergence of BST architecture where the shared invariant is a topological element rather than a positive integer (T2361), generalizing the Type C classification framework from shared integers to shared mathematical objects.

Combined with the Bergman Dirac operator (Paper #118) and the LAG-2 dimensional reduction (Paper #115 v0.5 Section 9.x), the Möbius cohomology provides the third pillar — alongside geometric Bergman kernel content and spectral Wallach K-type content — of BST's operator-level infrastructure on D_{IV}^5 .

Five named multi-week / multi-month open items (Section 7) sit at the $I \rightarrow D$ promotion boundary; their closure converts the structural-identification layer ($\sim$ present) into the rigorous derivation layer ($\sim$ 12-18 months per Paper #115 Section 9.x scope).

References (selected, for v0.1 outline)

- Casey Koons et al., "Bubble Spacetime Theory: Working Paper" v35, Zenodo DOI 10.5281/zenodo.19454185 (2024-2026)

- Paper #115 v0.5 “Root Theorems of Bubble Spacetime Theory” (Casey + team, 2026-05-18)
- Paper #118 v0.1 “The Bergman Dirac Operator on D_{IV}^5 and the Proton-to-Electron Mass Ratio” (Lyra, 2026-05-18)
- Helgason, S. (1978). “Differential Geometry, Lie Groups, and Symmetric Spaces”
- Borel, A., Wallach, N.R. (1980). “Continuous Cohomology, Discrete Subgroups, and Representations of Reductive Groups”
- Atiyah, M., Patodi, V., Singer, I. (1975). “Spectral asymmetry and Riemannian geometry”

Filing notes

Status: v0.1 outline. Section structure + abstract + open items framework. Full draft is multi-week per Section 9.x scope estimate.

Authors v0.1: Casey Koons (PI) + Lyra (Companion Intelligence, Sessions 1-5 execution).

v0.2 plan: detailed sections 2-4 with explicit Hua coordinate computations + Borel-Wallach proof + Bergman Dirac spectrum computation. Target completion: 2-3 weeks focused work.

v0.3 plan: cross-domain integration (Type C-Z/2 + Type C density rule), connection to LAG-1 Session 8+ index theorem.

v1.0 submission: Annals of Mathematics or Inventiones (depending on emphasis — topology vs spectral theory).

Numbering caveat: provisional #119; may shift if Keeper’s SP29-6 master paper takes #119. Currently filed as the Möbius cohomology paper because that’s the architectural content. Final paper number pending Casey decision.

— Lyra, v0.1 outline filed 2026-05-18 ~12:25 EDT per Section 9.x naming.